MA416

Lecture 01 – September 2, 2025
Quick Review on One-Sample Mean Testing

Dr. Matthew N. Moore

General Hypothesis Testing Methodology for Means

We start with the generic two-sided alternative:

$$H_0: \mu = \mu_0$$
$$H_0: \mu \neq \mu_0$$

and typically choose $\bar{X} \coloneqq \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$ as our estimator for μ, where:

$$X_k \sim^{i.i.d.} P(\mu, \sigma^2)$$

with P not necessarily normal. We know that (be able to prove):

$$\mathbb{E}(\bar{X}) = \cdots = \mu$$
$$\mathbb{V}(\bar{X}) = \cdots = \frac{\sigma^2}{n} \xrightarrow{n \to \infty} 0$$

Implying that $\bar{X}$ is both an unbiased and consistent estimator of μ. Note here also that:

$$\mathrm{SE}(\bar{X}) = \sqrt{\mathbb{V}(\bar{X})} = \frac{\sigma}{\sqrt{n}}$$

General Hypothesis Testing Methodology for Means

Since σ^2 is typically not known, we know that:

$$S^2 := \frac{1}{n-1} \cdot \sum_{k=1}^{n} (X_k - \bar{X})^2 \quad \rightarrow \quad \mathbb{E}(S^2) = \sigma^2$$

In addition, we have:

$$\mathbb{V}(S^2) = \cdots = \frac{2\sigma^4}{n-1} \xrightarrow{n \to \infty} 0$$

Implying that S^2 also is an unbiased and consistent estimator of σ^2. Therefore, in the case that σ^2 is unknown, we have:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{S}{\sqrt{n}}$$

Test Statistics for $H_0: \mu = \mu_0$

Also, from probability theory, one can verify that if $X_k \sim^{iid} N(\mu, \sigma^2)$, then:

$$\frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu_X}{\sigma_X/\sqrt{n}} \sim N(0,1) = Z$$

Therefore, if H_0 is correct, then $\mu_X = \mu_0$, and hence we choose:

$$Z_{stat} = \frac{\bar{X} - \mu_0}{\sigma_X/\sqrt{n}} \sim Z$$

In the case σ_X is unknown, also from probability one can verify that:

$$\frac{\bar{X} - \mu_X}{S_X/\sqrt{n}} \sim T_{n-1}$$

So, if H_0 is correct, we choose:

$$T_{stat} \sim \frac{\bar{X} - \mu_0}{S_X/\sqrt{n}} \sim T_{n-1}$$

Critical Values for $H_0: \mu = \mu_0$

One can conclude the hypothesis test by comparing the test statistics to the associated critical values. Recall that:

$$H_a: \mu \neq \mu_0, \sigma \text{ known} \quad \rightarrow \quad \begin{cases} \int_{z_{crit_R}}^{\infty} f_Z(x)\,dx = \frac{\alpha}{2} \\ \int_{-\infty}^{z_{crit_L}} f_Z(x)\,dx = \frac{\alpha}{2} \end{cases}$$

$$H_a: \mu \neq \mu_0, \sigma \text{ unknown} \quad \rightarrow \quad \begin{cases} \int_{t_{crit_R}}^{\infty} f_{T_{n-1}}(x)\,dx = \frac{\alpha}{2} \\ \int_{-\infty}^{t_{crit_L}} f_{T_{n-1}}(x)\,dx = \frac{\alpha}{2} \end{cases}$$

Decision rule:

- If $z_{stat} \in (z_{crit_L}, z_{crit_R})$, fail to reject H_0
- If $z_{stat} \notin (z_{crit_L}, z_{crit_R})$, reject H_0

R-Code Sample

```
alpha = 0.05
n = 23

zcritL = qnorm(alpha/2, lower.tail=T)
zcritR = qnorm(alpha/2, lower.tail=F)

tcritL = qt(alpha/2, n-1, lower.tail=T)
tcritR = qt(alpha/2, n-1, lower.tail=F)

zcritL
zcritR
tcritL
tcritR
```

```
> zcritL
[1] -1.959964
> zcritR
[1] 1.959964
> tcritL
[1] -2.073873
> tcritR
[1] 2.073873
```

Choice between Z and T

Note that if $P \neq N(\mu, \sigma^2)$, then:

- Z_{stat} does not follow a Z distribution
- T_{stat} does not follow a T_{n-1} distribution

The central limit theorem stats that if $n \to \infty$, then the distributions of Z_{stat} and T_{stat} both asymptotically approach a Z distribution. General rule of thumb with some preferred α value:

- If P is normal and σ is known, use Z

- If P is normal and σ is unknown, use T (regardless of n)

- If P is not normal but n is arbitrarily large:
  - If σ is known, use Z
  - If σ is unknown, use T (regardless of n)

- If P is not normal but n is not arbitrarily large, use non-parametric methods.

P-values for $H_0: \mu = \mu_0$

Once you have calculated your test statistic (either Z_{stat} or T_{stat}), a p-value can be calculated from it via

$$p_{val} = 2 \cdot \int_{|z_{stat}|}^{\infty} f_Z(x)\,dx \quad \text{or} \quad p_{val} = 2 \cdot \int_{|t_{stat}|}^{\infty} f_{T_{n-1}}(x)\,dx$$

Note that these only work for two-sided symmetric alternatives, H_a. If one-sided, drop the 2 term, and if non-symmetric, calculate the associated p-values and add.

Decision rule:

- If $p_{val} > \alpha$, fail to reject H_0
- If $p_{val} < \alpha$, reject H_0

R-Code Sample

```
# H_a: mu not equal to 170, sigma unknown

S = c(167, 178, 183, 184, 167)
xbar = mean(S)
sx = sd(S)
n = length(S)

tstat = (xbar - 170)/(sx/sqrt(n))

pval = 2*pt(abs(tstat), n-1, lower.tail=F)

pval
```

```
> pval
[1] 0.1952784
```

Confidence Intervals for $H_0: \mu = \mu_0$

We define the margin of error for a confidence interval to be:

$$\varepsilon = z_{crit} \cdot \mathrm{SE}(\bar{X})$$

(replace z_{crit} with t_{crit} as necessary) and define:

$$(CI)_{\mu}^{c\%} = (\bar{X} - \varepsilon, \bar{X} + \varepsilon)$$

where $c = 1 - \alpha$, which is called the two-sided symmetric confidence interval for μ based on the estimator $\bar{X}$.

Decision rule:

- If $\mu_0 \in (CI)_{\mu}^{c\%}$, fail to reject H_0
- If $\mu_0 \notin (CI)_{\mu}^{c\%}$, reject H_0

R-Code Sample

```
# H_a: mu not equal to 170, sigma unknown
S = c(167, 178, 183, 184, 167)
xbar = mean(S)
sx = sd(S)
n = length(S)

sexbar = sx/sqrt(n)
tcrit = qt(alpha/2, n-1, lower.tail=F)
eps = tcrit*sexbar
ci = c(xbar - eps, xbar + eps)

round(ci, 2)
```

```
> round(ci, 2)
[1] 165.43 186.17
```